

VIRTUAL ULTRASONIC PARKING SENSOR

A Linux kernel driver + C++ monitoring project

What is this project?

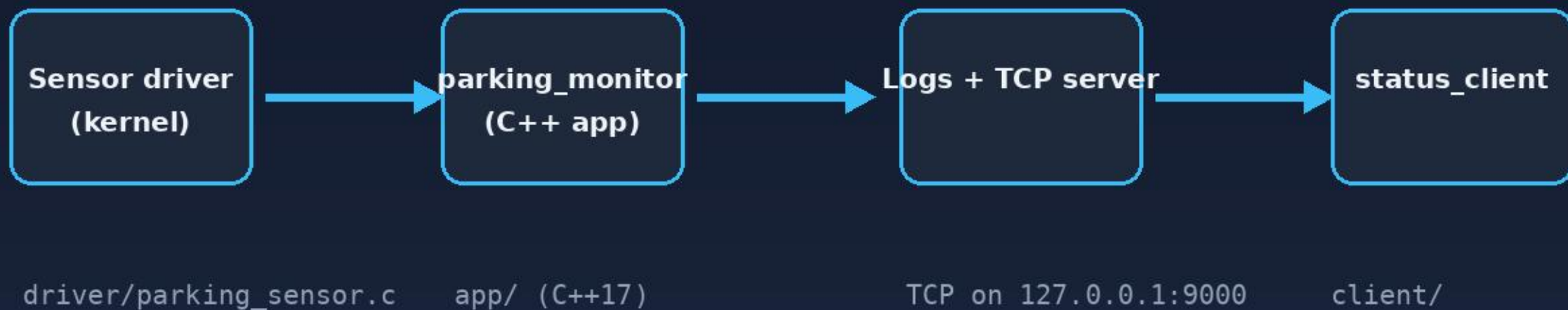
Simulates a car parking sensor in software (no hardware)

Kernel driver generates distance: 150 cm down to 10 cm and back

C++ app classifies each reading: SAFE / WARNING / DANGER

Logs everything and serves live status over TCP

System Architecture



Distance to State Rules

`distance > 50 cm`

SAFE

`21 to 50 cm`

WARNING

`distance <= 20 cm`

DANGER

Project Structure

driver/parking_sensor.c - kernel character device

app/ - monitor (config, sensor, state, logger, TCP)

client/ - status_client TCP reader

tests/ - 41 tests, 1140 checks

docs/, uml/, scripts/ - documentation & helpers

Design Highlights

Driver reports raw distance; policy decides SAFE/WARNING/DANGER

--simulate flag swaps in a virtual sensor - runs without root

Interfaces (ISensorSource, ThresholdPolicy) make it testable

Plain TCP on loopback, no external libraries

How to Build & Run

```
$ make
```

```
    builds parking_monitor, status_client, run_tests
```

```
$ make test
```

```
1140 checks run, 0 failed
```

```
$ ./build/parking_monitor --simulate --iterations 20
```

```
STATE CHANGE -> SAFE at 140 cm
```

```
$ ./build/status_client --port 9000 --count 3
```

```
STATUS distance=37 state=WARNING
```

Testing & Quality

Unit tests: thresholds, config parsing, logger, telemetry

Integration: full run writes every state to the log

Live TCP server/client check in test_all.sh

Final result: 1140 checks, 0 failures

Conclusion

Demonstrates Linux device drivers, threads, sockets, file I/O

Clean layered, interface-driven C++ design

Fully tested and documented, runs with or without root

GitHub: github.com/rajarp3108-tech/Virtual_Sonic_Parking